



The Assignment Guidelines

You must choose **ONE** of the following three projects. The goal is to build a working, end-to-end system—not just UI screens.

- **Preferred Tech Stack:** React (You are free to use any modern full-stack ecosystem).
- **Evaluation Focus:** System design thinking, frontend/backend architecture, AI integration, scalable structure, and clean UI/UX.

Option 1: AI Social Media Content Pipeline

Problem Statement: Build a platform that helps businesses or creators generate complete social media content pipelines using AI. The system should allow a user to create a “brand/project workspace” where they can generate social media content such as:

1. Instagram captions
2. LinkedIn posts
3. Twitter/X threads
4. hashtags
5. carousel text
6. short marketing content
7. campaign ideas

The platform should use Google Gemini APIs for text generation and image generation. The idea is to simulate a lightweight AI-powered social media marketing SaaS product.

Minimum Requirements:

1. Authentication
 - a. User signup/login
 - b. Protected dashboard
2. Workspace System
 - a. Create multiple projects/workspaces
 - b. Store generated content history
3. AI Content Generation

User should be able to:

 - a. enter brand/business details
 - b. select platform (Instagram, LinkedIn, Twitter/X etc.)
 - c. select tone/style
 - d. generate content using Gemini
4. AI Image Generation

Generate accompanying social media images/posters using Gemini image generation.

5. Content Management
 - a. Save generated content
 - b. Edit generated content
 - c. Delete/regenerate content
6. Scheduling UI

Create a UI for scheduling posts (actual posting not necessary).
7. Export System

Allow export as:

 - a. PDF
 - b. Markdown
 - c. ZIP/JSON
8. Bonus Features (Optional but Valuable)
 - a. AI-generated reel/video scripts
 - b. Multi-image carousel generation
 - c. Team collaboration
 - d. Real-time AI streaming responses
 - e. AI content approval workflow
 - f. Calendar view
 - g. Prompt templates
 - h. Dark/light mode
 - i. Analytics dashboard

What We Are Evaluating

- API architecture
- frontend state management
- AI integration quality
- scalability thinking
- clean UI/UX
- database design
- code organization
- async handling

Helpful Hints

- Use Gemini text models for captions/content generation.
- Use Gemini image generation for posters/banners.
- Store prompts and responses in the database.
- Use reusable prompt templates instead of hardcoded prompts.
- Structure backend services separately:
 - auth
 - AI services
 - workspace management
 - exports
- Think carefully about loading states and error handling.

- Focus on making the system extensible.

Option 2: AI Website & Content Builder for Businesses

Problem Statement: Build an AI-powered website generation platform for small businesses.

The platform should allow a user to enter business information such as:

- business name
- category/industry
- services
- target audience
- preferred tone/style

Based on this information, the system should generate:

- website content
- landing page sections
- FAQs
- testimonials
- contact information blocks
- SEO content
- AI-generated images

The user should then be able to edit and customize the generated website inside the platform.

This project simulates a lightweight AI website builder product.

Minimum Requirements

1. Authentication
 - a. Login/signup system
2. Website Generator

User inputs business details and AI generates:

 - a. Hero section
 - b. About section
 - c. Services section
 - d. FAQ section
 - e. CTA section
3. AI Image Generation

Generate images/banner graphics relevant to the business.
4. Website Editor

Allow editing of:

 - a. text
 - b. sections
 - c. layout components
5. Live Preview

Real-time website preview while editing.
6. Project Saving

Users should be able to:

- a. save projects
 - b. reopen projects
 - c. continue editing later
7. Bonus Features (Optional but Valuable)
 - a. Drag-and-drop editor
 - b. AI-generated logos
 - c. Theme/color generator
 - d. One-click deploy
 - e. Export static website
 - f. Mobile responsive preview
 - g. AI chatbot assistant inside editor
 - h. Regenerate only selected sections

What We Are Evaluating

- dynamic frontend rendering
- component architecture
- AI workflow design
- editor state management
- backend structure
- database schema quality
- user experience

Helpful Hints

- Treat website sections as dynamic JSON components.
- Store page layouts in structured format.
- Use modular frontend components.
- AI output should be editable, not static.
- Avoid generating plain text blobs — structure responses cleanly.
- Think about how a no-code editor internally works.
- Consider separating:
 - AI generation layer
 - rendering layer
 - editor layer

Option 3: AI Reel Script & Thumbnail Generator

Problem Statement:

Build a platform for content creators that helps generate short-form social media content using AI.

The platform should help users generate:

- reel/video ideas
- viral hooks
- scripts
- scene breakdowns

- captions
- hashtags
- thumbnails/posters

The system should use Gemini APIs for both text generation and image generation.

This project simulates an AI-powered creator assistant platform.

Minimum Requirements

1. Authentication
 - a. User login/signup
2. AI Script Generation

User enters:

 - a. topic
 - b. niche/category
 - c. platform
 - d. content style
3. AI should generate:
 - a. video title
 - b. hook
 - c. short-form script
 - d. scene structure
 - e. CTA
 - f. hashtags
4. Thumbnail Generation

Generate AI thumbnails/posters using Gemini image generation.
5. Script Management
 - a. Save scripts
 - b. Edit scripts
 - c. Duplicate scripts
 - d. Organize by folders/projects
6. Content Dashboard

Display all generated content in a clean dashboard.
7. Bonus Features (Optional but Valuable)
 - a. Viral score prediction
 - b. AI storyboard generation
 - c. Auto subtitle generation
 - d. Voice synthesis integration
 - e. Multi-platform optimization
 - f. Script-to-carousel conversion
 - g. Trending topic suggestions
 - h. Real-time collaboration

What We Are Evaluating

- AI prompt engineering
- frontend polish

- backend quality
- project architecture
- creativity
- product thinking
- usability

Helpful Hints

- Break script generation into multiple AI steps instead of one huge prompt.
 - Use separate prompts for:
 - hooks
 - scripts
 - hashtags
 - thumbnails
 - Use queues/background jobs for image generation.
 - Organize generated content properly in DB collections/tables.
 - Think about creator workflow and usability.
 - Handle AI failures gracefully.
 - Strong UX matters in this project.
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The objective of these assignments is not just to test coding ability, but to evaluate:

- system design thinking
- frontend/backend engineering
- AI integration skills
- scalability approach
- problem-solving ability
- product understanding
- code quality and architecture